



Microtek E 625 UPS

ing and ignition of any automotive application. As soon the automobile starts, the battery becomes idle. Inverters, on the other hand, require a steady current for a long duration.

So use only inverter-grade batteries with inverters for better results and long life.

Auto-adjustment to battery

Not all the inverters offer the facility for auto-adjustment to different types of batteries (standard, tubular and SMF). If auto-sensing isn't there, the battery will get under-/over-charged, which will affect the backup time as well as battery life.

Battery charging characteristics are measured on two counts: boost voltage and float voltage. These settings should change according to the type of battery used. HI-LITE inverters of UTL come with this auto-adjustment feature.

Some manufacturers offer an intelligent charging profile adjustment wherein the inverter can sense the charging profile need of a battery and adjust the charging accordingly. To give you an example, local batteries require a different kind of charging than maybe a branded battery. When a company makes an inverter, it doesn't know whether a branded or a local battery will be used with the inverter. So if the charger in the inverter is set for a local battery and you put a branded battery, then it's a problem. These inverters automatically sense after a few cycles of charging as to

what's the type of battery and what's the best charging profile required and perform the required adjustment in time.

In the event of a very high or low voltage, the inverter cuts off and switches to the battery. A very low voltage affects the charging. So some inverters are designed for charging from as low as 110V, keeping in mind the varied power conditions in India.

Price

Home inverter requirement can go up to as high as 40 kVA for farmhouses. But 600VA and 800VA are the largest selling models, accounting for 85 per cent of the market. 1.5 and 3kVA inverters hold about 12-13 per cent share and the balance is accounted for by 10kVA and higher models.

Microtek's 600VA and 850VA inverters start at Rs 3990 and Rs 4990, respectively. The average price for Luminous 600VA and 800VA inverters is Rs 3500 and Rs 4500, respectively. UTL's 800VA inverters cost Rs 4500 for squarewave and Rs 5200 for sinewave versions. Parker Power's 630VA and 860VA Digital Home UPS start at around Rs 3300 and Rs 4300, respectively.

Last but not the least

Buy only a branded inverter as there are many down-the-line suppliers who just assemble inverters from the kits available in the market. These local

Checklist

- Built-in UPS
- Squarewave, modified sinewave or pure sinewave
- Size of the inverter
- Backup duration/battery size
- Type of battery (standard, tubular or SMF)
- Same battery and inverter brands
- Protection features
- Innovative features like auto-adjustment to battery, multiple charging, etc
- After-sales service
- Price

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type of companies offer very low price but the reliability of their inverter is almost nil as they do not have any manufacturing or R&D facility. See that you pay a reasonable price for the branded inverters and check the warranty terms and period of warranty.

Apart from this, after-sales service is very important and many manufacturers lag in that.

The next important factor is the battery of the inverter. Many companies sell only inverter and one has to buy the battery from another source. Complications arise when the inverter or battery fails. Get inverter and battery from the same vendor.

Also ensure that the inverter has features of multiple charging, which means that it has regulation for fast charging, normal charging, branded batteries and so on.

Other points to consider are features like child lock, compatibility with solar panel and input voltage range. ●

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